

Far-space neglect in conjunction but not feature search following transcranial magnetic stimulation over right posterior parietal cortex

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¹*Institute of Cognitive Neuroscience, National Central University, Zhongli, Taiwan;* ²*Institute of Neuroscience, National Yang-Ming University, Taipei, Taiwan;* ³*Laboratories for Cognitive Neuroscience, National Yang-Ming University, Taipei, Taiwan;* ⁴*Institute of Linguistics, Academia Sinica, Taipei, Taiwan;* ⁵*Institute of Cognitive Neuroscience, University College London, London, United Kingdom;* and ⁶*Department of Psychology, Goldsmiths, University of London, New Cross, London, United Kingdom*

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Mahayana IT, Liu CL, Chang CF, Hung DL, Tzeng OJ, Juan CH, Muggleton NG. Far-space neglect in conjunction but not feature search following transcranial magnetic stimulation over right posterior parietal cortex. *J Neurophysiol* 111: 705–714, 2014. First published November 20, 2013; doi:10.1152/jn.00492.2013.—Near- and far-space coding in the human brain is a dynamic process. Areas in dorsal, as well as ventral visual association cortex, including right posterior parietal cortex (rPPC), right frontal eye field (rFEF), and right ventral occipital cortex (rVO), have been shown to be important in visuospatial processing, but the involvement of these areas when the information is in near or far space remains unclear. There is a need for investigations of these representations to help explain the pathophysiology of hemispatial neglect, and the role of near and far space is crucial to this. We used a conjunction visual search task using an elliptical array to investigate the effects of transcranial magnetic stimulation delivered over rFEF, rPPC, and rVO on the processing of targets in near and far space and at a range of horizontal eccentricities. As in previous studies, we found that rVO was involved in far-space search, and rFEF was involved regardless of the distance to the array. It was found that rPPC was involved in search only in far space, with a neglect-like effect when the target was located in the most eccentric locations. No effects were seen for any site for a feature search task. As the search arrays had higher predictability with respect to target location than is often the case, these data may form a basis for clarifying both the role of PPC in visual search and its contribution to neglect, as well as the importance of near and far space in these.

near space; far space; parietal cortex; visual search

THE ROLE OF THE POSTERIOR parietal cortex (PPC) in spatial processing has been the subject of study for a number of years, with research relating to reaching and visuomotor behavior indicating that it is involved in attending to near space or the space within manipulatory range rather than far space (Vuilleumier et al. 1998). On the other hand, some studies have shown that in hemispatial neglect, a condition resulting from damage to, amongst other areas, the parietal cortex, the impairment is observed more severely in far space (“far neglect”) (Butler et al. 2004; Cowey et al. 1994, 1999). This suggests that the PPC is not limited to a role in near-space spatial processing, also having a role in far-space spatial control and with most cases of neglect, due to parietal lobe damage (Butler et al. 2004; Colby and Goldberg 1999; Corbetta et al. 2005;

Corbetta and Shulman 2011; Cowey et al. 1994; Mesulam 1999). Furthermore, whereas spatial information is coded predominantly symmetrically in visual areas, this is not the case in the parietal cortex, where it is represented asymmetrically (Sommer et al. 2008) and emphasizes the complexity of spatial representations in the human brain.

Near and far spatial functions are represented in dorsal and ventral divisions of the visual association cortex (Halligan and Marshall 1991; Pitzalis et al. 2001). As part of the dorsal network, the right PPC (rPPC) connects with, amongst others, right frontal eye fields (rFEFs) and carries out visuospatial processing functions (He et al. 2007; O’Shea et al. 2006). The rFEF has been argued to be involved in shifting attention, but research has not indicated specificity for near or far spatial processing typically (Grosbras and Paus 2002; Muggleton et al. 2003, 2010). On the other hand, an area within the ventral visual stream—the right ventral occipital cortex (rVO)—has been identified as important, specifically for far space (Weiss et al. 2000). Therefore, in this study, in addition to investigating rPPC, rFEF and rVO were studied, in line with work in several near- and far-space studies (Berti et al. 2002; Lane et al. 2013) and allowing for comparison with rPPC function. Additionally, few studies have investigated the importance of rPPC in far spatial processing using repetitive transcranial magnetic stimulation (TMS). The current study aimed to use repetitive TMS to investigate the role of rPPC in neglect in both near and far space and used an elliptical array of conjunction visual search stimuli. The use of such an array to assess performance allows for a clearer assessment of the effects of the location of the target than the typical random search array used in visual search studies, including the recent investigation by Lane et al. (2013). In their study, the search performance, with respect to the laterality of target location, was not reported. As visuospatial neglect is characterized by a lack of awareness for sensory events located toward the contralesional side of space (Driver and Mattingley 1998; Driver and Vuilleumier 2001), the nature of any effects with respect to laterality still requires investigation. Additionally, eye-movement monitoring or elimination from the experimental paradigm is important to control for effects that could occur as a consequence of these being affected.

TMS has been used to help assess the involvement of several brain areas in visual search (Walsh et al. 1998), impairments that are also observed in visuospatial-neglect patients

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(Behrmann et al. 1997; Kristjansson et al. 2005), who have difficulty attending to or fail to respond to contralesional stimuli (Azouvi et al. 2006; Deouell et al. 2000; Driver and Mattingley 1998; Hillis 2006). The presence of neglect is an important predictor of poor functional recovery in stroke patients, and it may greatly restrict the daily activities of patients (Jehkonen et al. 2000). Greater knowledge of the specifics of the contribution of PPC to neglect may provide insights into healthy brain function, help predict long-term outcomes in patients (Gialanella and Ferlucci 2010), and help provide a basis to guide brain stimulation-based therapies (Ting et al. 2011).

The current study used a visual search paradigm to assess the neglect-like pattern in normal individuals. In patients with rPPC damage (showing left neglect), they tend to make fewer fixations and have shorter inspection times for items on the contralesional left side of space (Behrmann et al. 1997); hence, the use of a structured search array and visual search performance has been found to be a more sensitive indicator of distance-related changes in performance in neglect than tasks, such as line bisection (Aimola et al. 2012). In relation to spatial functions, Butler et al. (2009) found that neglect patients showed a decrease in the proportion of target detections from right to left in both near and far space. This type of target identification requires the parietal cortex, which has been argued to perform functions, such as feature integration and the perceptual analyses required for target identification (Nobre et al. 2003). As such, rPPC is essential for conjunction search and has a key role in both landmark and hard conjunction tasks (Ellison et al. 2004). Although rPPC seems not essential for feature search (Corbetta et al. 1995; Ellison et al. 2003), the overlapping frontoparietal networks, which include the PPC and FEF, are engaged by both feature and conjunction search (Donner et al. 2002), and the FEF is essential for visual search, even in the absence of a requirement for eye movements (Muggleton et al. 2003). Additionally, during the processing of targets and distractors in search, other areas, such as the VO, also show activation (Kim and Hopfinger 2010).

The aim of this present study was to assess the presence of a left-right performance difference (neglect) in normal individuals in near space and far space using a visual search task with manual responses as a consequence of TMS. The elliptical conjunction search used contained elements in the peripheral visual field, with a range of horizontal offsets from the center. This design was used as a consequence of neglect patients showing a gradual reduction of perception across space in one or more dimensions (Dvorkin et al. 2011) and prior research on neglect having typically focused on only one dimension of space, either defining deficits in horizontal dimensions (Giglia et al. 2011; Gobel et al. 2006; Sack 2010) or radial dimensions (Berti and Frassinetti 2000; Lane et al. 2013; Vuilleumier et al. 1998; Weiss et al. 2000) separately. It was hypothesized that the visual search task, using peripheral locations, would allow the dissociation (if any) of rPPC neglect-like patterns for the near- or far-space reference frames.

MATERIALS AND METHODS

Subjects

For the first experiment, 12 subjects (eight men and four women; mean age: 24 ± 2.9 SD), with no previous history of neurological

problems, participated in the experiment. All participants had normal or corrected-to-normal vision, were right handed, and were not colorblind. All participants had participated in TMS experiments previously but were naive with respect to the purpose of the experimental procedures. For the second experiment, 12 subjects were recruited (six women and six men; mean age: 23.7 ± 2.3 SD), four of whom had participated in the first experiment (two women and two men; mean age: 25.5 ± 1 SD). All subjects gave written, informed consent, and the local Research Ethics Committee approved the study.

Apparatus

All stimuli were presented on a 19-in. cathode ray tube monitor with a 75-Hz refresh rate driven by an Intel Core i7 personal computer and were programmed with E-Prime (Psychology Software Tools, Sharpsburg, PA). Subjects were seated comfortably, either 70 or 140 cm away from the screen, with the center of the screen at eye level for the near and far conditions, respectively. Head position was controlled by a chinrest.

Each participant's head was coregistered with his/her own T1 MRI brainscan usingBrainsight frameless stereotaxic software (Rogue Research, Montreal, Quebec, Canada) to confirm the anatomical locus of stimulation. A Super Rapid stimulator (Magstim, Whitland, UK) was used to deliver the magnetic stimulation. Eye movements from the subjects were recorded with an EyeLink 1000 infrared video system (SensoMotoric Instruments, Teltow, Germany).

Stimuli and Design

Experiment 1. The search array consisted of 12 items, in which six stimuli were positioned in each visual field (Fig. 1A). Then, the six locations in each visual field were divided equally into three locations horizontally, with a pair of stimuli at each: one in the upper visual field and one in the lower visual field, with distances from the midline of 2° , 4.5° , and 6.5° . This allowed for analysis of the results in terms of whether it was influenced by the position of the stimulus laterally or according to the stimulus eccentricity (Cazzoli et al. 2009).

The vertical distance of stimuli locations was shorter than the horizontal distance from fixation to make the array into an ellipse. A study by Carrasco et al. (2001) found performance asymmetry for vertical and horizontal offsets. Therefore, in the present study, the vertical distance maximum was 5.6° , and the horizontal distance maximum was 6.5° from the fixation. These distances are similar to those of Silver et al. (2007), who used a visual detection task with a size of 7° by 5° visual. The sizes of the stimulus displays were set in both the near and far conditions, such that they were the same in terms of degrees of visual angle.

The search stimuli were backward diagonals (\) or forward diagonals (/) and either red (R) or green (G). The target was a/G, and the distractors were /R and /G diagonals (Fig. 1B). Each stimulus in the visual search array was 1° by 1° (1.3° in length), based on studies estimating the attentional focal point to be 1.5° in angle (Humphreys 1981; Thompson et al. 2005), although it varies depending on the location of the stimulus [0.8° by 0.6° in foveal visual field and 1.5° by 1.1° in the periphery (Giesbrecht et al. 2003)].

For the definition of near and far space, the study used values based on previous research; for example, near space is defined as a distance within reaching distance (Weiss et al. 2000; Woodin and Allport 1998). For far-space distance, studies have used a variety of distances. For example, a study from Mennemeier et al. (1992) used distances >60 cm from the subjects; Berti et al. (2002) used 100 cm; Pitzalis et al. (2001) defined far space as 160 cm; and Lane et al. (2013) defined it at a distance of 172 cm. In the current study, the distance was fixed at 70 cm away from the participant for near space and 140 cm for far space (Fig. 2B).

Experiment 2. The second experiment was the same as the first, except for the elements of the search array. This time, the task was a

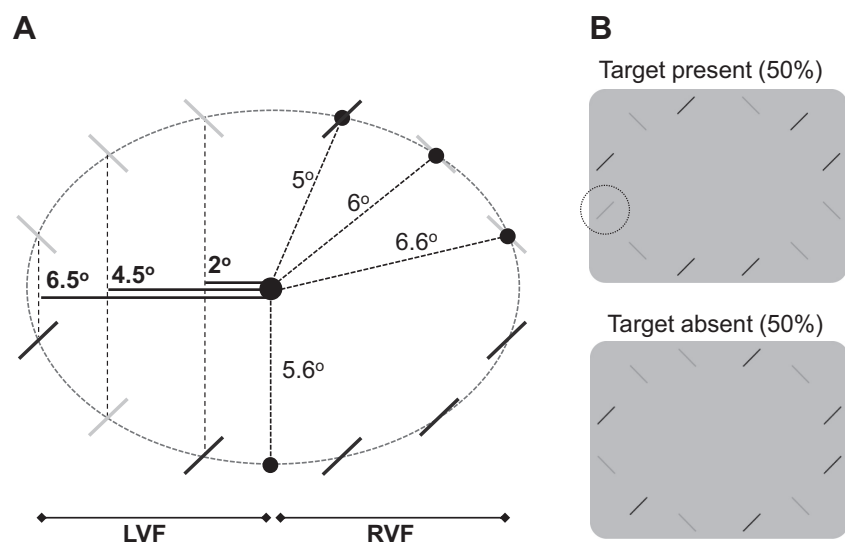


Fig. 1. A: target/distractor size and stimulus display. All 12 target/distractor locations were positioned in an elliptical configuration. B: the conjunction search stimuli displayed in the experiment for the target-absent and target-present conditions (darker gray: red; lighter gray: green). LVF, left visual field; RVF, right visual field.

feature search task, and the target was a green circle and the distractors red circles. The size of these matched the size of the elements in experiment 1.

Procedure

Experiment 1. Each experimental block consisted of 48 trials to avoid TMS coil overheating (Sommer et al. 2006), which consisted of 24 target-present and 24 target-absent trials. A trial consisted of a fixation cross (400 ms), followed by presentation of the search array for a duration determined for each participant, which was then masked immediately until a response was made using 12 pattern masks with the same dimensions as the elements of the search array (Fig. 2A). Participants were required to indicate the presence or absence of a target by a key press using their right-hand first finger (present) and

index finger (absent) to make their responses. The target occurred at an equal frequency in each of the 12 array locations in a random order, and the presence or absence of the target was also randomized.

The duration of presentation of the stimulus was determined for each participant using a Bayesian adaptive thresholding method, which had been used in previous studies (Kalla et al. 2008). Throughout the course of a block of 48 trials, the algorithm appropriately adjusted the presentation duration to result in an estimate of the duration required to produce performance with 75% accuracy, roughly equivalent to a sensitivity index (d') of one (Kontsevich and Tyler 1999).

Each subject completed two sessions (one for near and one for far space). These were carried out on separate days with at least 7 days between sessions. Near- and far-space sessions were carried out in a counterbalanced order. All blocks in a session were run in random

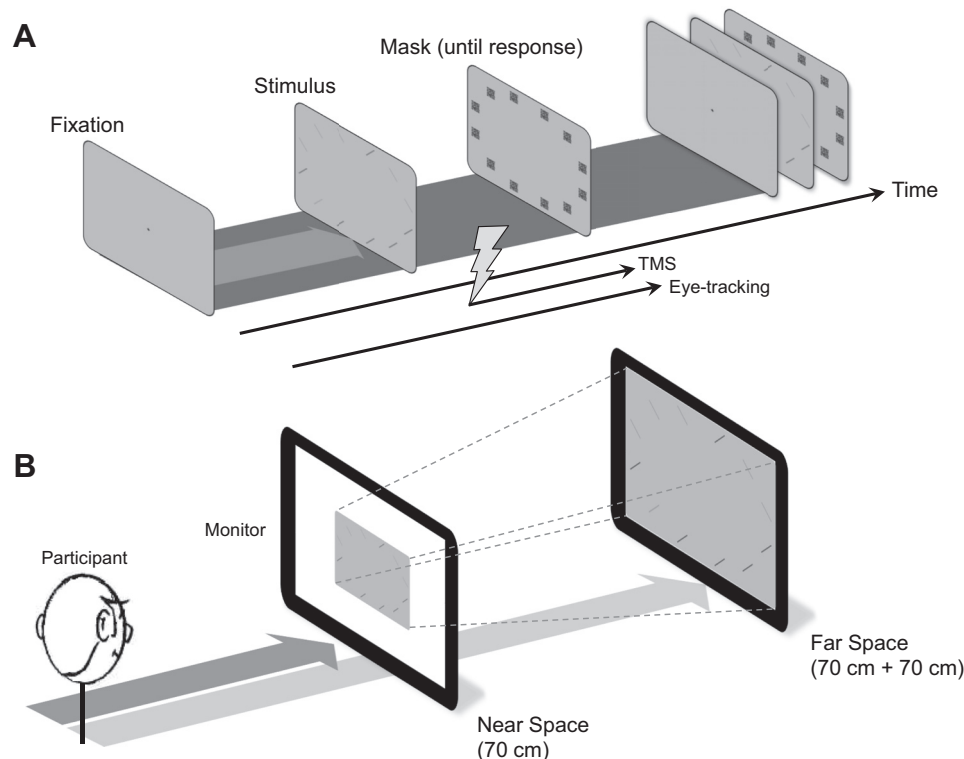


Fig. 2. A: stimulation procedure. The stimulation procedure started with a fixation (400 ms) and was followed by the stimulus display (individually determined duration, based on individual thresholds) and then the mask (until response). Five pulses (10 Hz, 500 ms, 60% intensity transcranial magnetic stimulation (TMS), were delivered at the onset of the stimuli display. Eye tracking took place from fixation onset until the end of the search array presentation to monitor events, such as blinks and saccades. B: near and far distance parameters. In the near-space condition, the subject sat 70 cm from the monitor and in the far-space condition, 140 cm from the monitor.

order. For each session, participants completed three steps consisting of a practice block, thresholding blocks, and formal test blocks. The practice block familiarized subjects with the search task. The formal test blocks were started following three thresholding blocks, which were used to get the appropriate duration of the stimulus display for the subject, with the lowest duration selected from the three obtained and used in the formal test blocks. The formal test blocks consisted of eight blocks, which contain two blocks of rFEF stimulation, two blocks of rPPC stimulation, two blocks of rVO stimulation, and two blocks of vertex stimulation. The vertex-stimulation site functioned as a control condition. All blocks in one session were run in random order, and the near and far condition sessions were run in a counter-balanced order.

In the formal right TMS (rTMS) blocks, the TMS pulses were delivered concurrent with the onset of the visual search array.

Eye Monitoring

Due to the relative difficulty of the search task used, it was expected that array durations would have a high likelihood of being >200 ms for some subjects, a duration consistent with saccade onset latencies in this sort of task (Chelazzi et al. 1993; Munoz and Wurtz 1992) (this occurred for seven subjects). Therefore, to ensure that the data were not confounded by saccades (or blinks), eye position was monitored during the experiment. This was achieved using the Eye-Link 1000 (SensoMotoric Instruments), which recorded eye-position information during intervals defined by the E-Prime program. The eyes were monitored from the onset of the fixation cross until the presentation of the mask. On presentation of the fixation point, the program waited for the subject's eyes to fixate on the fixation cross. If the fixation was achieved, the trial was run. If no fixation was recorded within 5 s, the trial was counted as invalid and excluded from analysis. In addition, during the stimuli display, if any blinks or eye movements were detected, then the trial was also excluded, and any response data discarded. Across all subjects, a total of 135 trials was excluded, due to eye movements or blinks, equivalent to 11.25 trials/subject or 1.17% of the total trials.

Experiment 2. The procedure for *experiment 2* was the same as that for *experiment 1*.

rTMS Procedure and Site Localization

rTMS procedure. Repetitive TMS was delivered during experimental sessions using a Super Rapid stimulator (Magstim) connected to a 70-mm figure-of-eight coil for the rPPC condition and to a 55-mm figure-of-eight coil in the rFEF and rVO conditions (to minimize twitches), as well as the vertex stimulation. Each event-related train of rTMS comprised a 10-Hz, five-pulse delivery (500 ms in duration). A fixed stimulus intensity of 60% of machine output was used, as previous studies have shown both that TMS is able to evoke clear responses at relatively low intensities, such as 60% of stimulus intensity (Komssi et al. 2004), and that there is little correlation between thresholds for different brain areas that might otherwise guide stimulation settings (Stewart et al. 2001). In the absence of an ideal method for selecting stimulus intensity, we chose 60%, as although it is likely to be below the motor threshold of some of the subjects, it has been shown to be effective in a number of previous studies and is at a level at which nonspecific effects, such as twitches or blinks, are likely to be relatively low. Coils were cooled before use to prevent overheating during a block and were replaced at the end of each block. Over rFEF and vertex, the coil was oriented parallel to the floor, with the handle running in an anteroposterior direction (Kalla et al. 2009). Over rPPC and rVO, the TMS coil was placed perpendicular to the anterior and posterior axis of the head and the handle at ~45° to the midline, pointing in a medial-to-lateral direction (Kalla et al. 2008).

Site localization. Several steps were performed to localize the sites for stimulation. For all subjects, a T1 anatomical brain MRI had been obtained previously. It was used in conjunction withBrainsight neuronavigation software to set the scalp coordinates of the stimulation site. This method greatly improves the anatomical localization before the TMS session (Nixon et al. 2004). Coordinates (Fig. 3), taken from previous studies [rFEF, Montreal Neurological Institute (MNI) coordinate: 31/−2/47 (Paus 1996); rVO, MNI coordinate: 28/−95/52 (Lane et al. 2013; Weiss et al. 2000); and rPPC (Talairach coordinate): 42/−58/52 (Bjoertomt et al. 2002; Muggleton et al. 2006)], were transformed into the Brainsight coordinate system using the Functional MRI of the Brain Software Library package (University of Oxford, Oxford, UK) and entered directly to the Brainsight software. Subjects were then coregistered with their structural scans, using readily identifiable points on the head (nose bridge and tip, intertragal notches of the ears), and the points on the surface of the head overlying the sites of interest were marked on a Lycra swimming cap worn by the subjects. These marks were used for coil positioning rather than tracking the coil position during TMS in the experiment using Brainsight, which might be considered a limitation of the study. However, sufficient care, when carrying out the stimulation, should minimize the effect of this on the data collected.

Experiment 2. The procedure for *experiment 2* was the same as that for *experiment 1*.

Data Analysis

In *experiments 1* and *2*, to compare the effect of TMS in near and far viewing distances, subjects' overall performance was measured using d-prime scores and mean correct reaction times (RTs) of target-present trials for rFEF, rPPC, rVO, and vertex-stimulation sites. The mean RTs were analyzed by excluding the individual RT outliers, where outliers were defined as outside mean ± 2 SD. No significant skewness was seen in the RT data. d-Prime represents the difference between the transformed hit and false-alarm rates and provides a good description of the relation between these when response bias varies (Macmillan and Creelman 2005). The d-prime and RT analyses followed steps of statistical analyses of a within-subject, repeated-measures, two-way ANOVA, subjected to a 4 (site: rFEF, rPPC, rVO, and vertex) $\times$ 2 (viewing distance: near and far) design to observe the interaction and each condition's main effects. This was followed by Bonferroni adjustment for multiple comparisons to test the different between-stimulation sites using paired-sample *t*-tests.

Additionally, in *experiment 1*, to test whether target location interacted with any neglect-like effect induced by TMS, accuracy and mean correct target-present RTs were calculated for the three different eccentricities (2°, 4.5°, and 6.5°) in the left and right hemifields. The hemifield analyses (the left vs. right visual fields) were done on all stimulation sites for each viewing distance (either near or far space). The statistical analyses steps were similar to the d-prime and overall RT analyses, except that the ANOVAs also included each viewing distance, resulting in a repeated-measures, three-way ANOVA, with factors of stimulation sites (rFEF, rPPC, rVO, and vertex), hemifield (right and left), and eccentricity (2°, 4.5°, and 6.5°).

RESULTS

Experiment 1

d-Prime scores were analyzed to assess the effects of TMS over all stimulation sites in the near- and far-space viewing-distance conditions (Fig. 4A). A two-way, repeated-measures ANOVA with a within-subject design revealed a significant interaction of stimulation sites (rFEF, rPPC, rVO, and vertex) and viewing distances [near and far; $F(3,33) = 4.559$, $P = 0.009$], with a significant main effect of stimulation site [$F(3,33) = 9.569$, $P < 0.001$] and a trend (i.e., $0.05 < P < 0.1$)

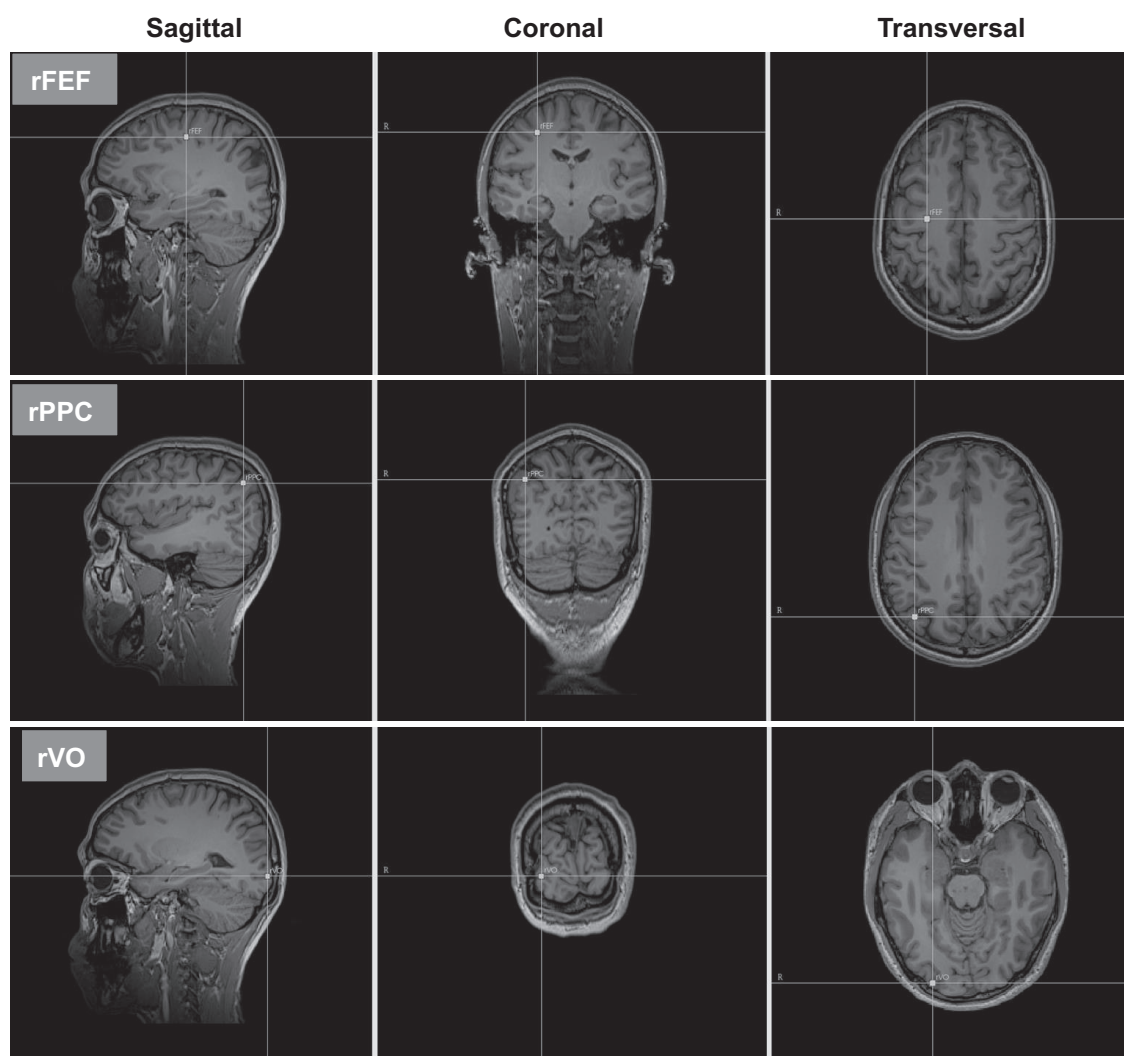


Fig. 3. TMS site localization. The right frontal eye field (rFEF), right posterior parietal cortex (rPPC), and right ventral occipital cortex (rVO) sites were localized using theBrainsight TMS-MRI coregistration system [Montreal Neurological Institute coordinates, rFEF: 31/−2/47 (Paus 1996); rVO: 28/−95/52 (Lane et al. 2013; Weiss et al. 2000); and Talairach coordinate, rPPC: 42/−58/52 (Bjoertomt et al. 2002; Muggleton et al. 2006)]. The rPPC coordinate lay in the region of angular gyrus lateral to the intraparietal sulcus. To localize vertex, we used anatomical localization, measuring the point on the midway between the intertragal notches and midway between nasion andinion.

for the viewing-distance condition [$F(1,11) = 3.990$, $P = 0.071$]. Bonferroni adjusted post hoc comparisons showed a significant difference of the rFEF, rPPC, and rVO from the vertex condition ($P = 0.005$, $P = 0.042$, and $P = 0.007$, respectively). This was supported by a significantly worse performance in far space compared with near space for rPPC [$t(11) = 2.589$, $P = 0.025$] but not in the rFEF [$t(11) = 0.704$, $P = 0.496$] and vertex [$t(11) = 1.686$, $P = 0.120$]. The worst performance for far-space performance with rPPC stimulation was similar to a decrease in performance for rVO in the far space compared with near space [$t(11) = 2.237$, $P = 0.047$].

To assess the TMS effects on target detection RT, we then analyzed the RT data from target-present trials (Fig. 4B). With the use of a two-way, repeated-measures ANOVA with a within-subject design, we found a trend ($0.1 < P < 0.05$) for interaction between stimulation site and viewing distance [$F(4,44) = 2.600$, $P = 0.069$], but there were no significant effects seen in the post hoc analyses. Paired-sample t -tests showed significant effects for near- vs. far-space conditions in the rPPC and rVO [$t(11) = -2.740$, $P = 0.019$, and $t(11) =$

-2.290 , $P = 0.043$, respectively], similar patterns to the d-prime analysis results.

Hemifield Analysis

An analysis of left vs. right visual fields was performed to assess the effects of TMS in terms of production of a “neglect-like” effect for the peripheral target locations in the conjunction visual search. In the near-space condition, the three-way ANOVA analysis with factors of stimulation site, hemifield, and visual field for the accuracy analysis revealed no significant main effects or interactions. The RT analyses revealed a significant interaction of stimulation sites and visual field [$F(3,33) = 3.416$, $P = 0.026$] and a significant main effect of eccentricity [$F(2,22) = 5.821$, $P = 0.009$]. Bonferroni adjusted pairwise comparisons of eccentricity showed that there was a significant difference between 2° and 4.5° eccentricity ($P = 0.003$). There were no significant results observed in the post hoc tests for right vs. left visual field in any stimulation sites at each eccentricity.

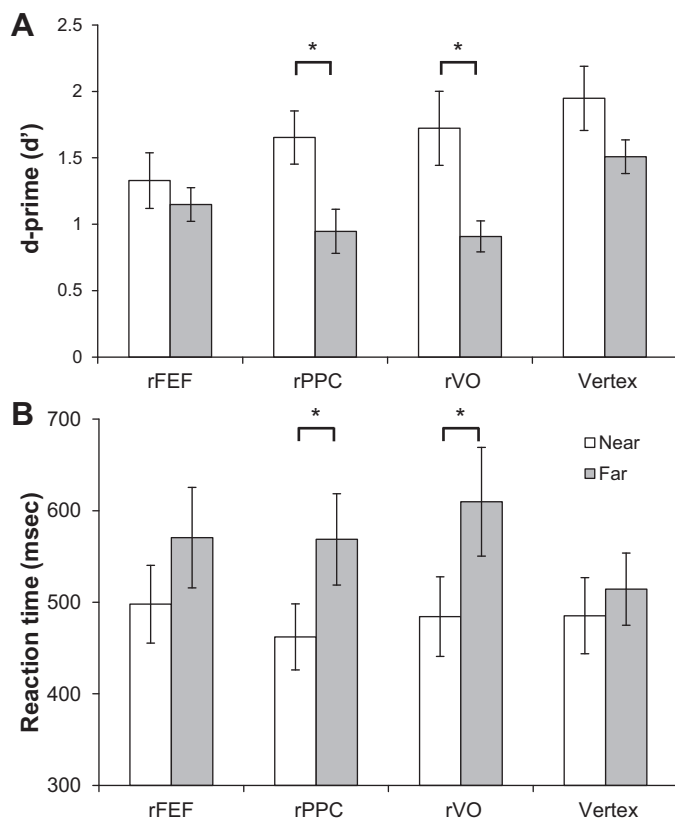


Fig. 4. Participants' performance across all stimulation sites in near and far space (experiment 1). A: sensitivity index (d-prime) scores. B: mean correct target-present reaction time (RT). * $P < 0.05$; error bars: SE.

In the far-space condition, the effects of TMS on performance across the two visual fields were more profound. The three-way ANOVA for accuracy analysis revealed a significant main effect of eccentricity [$F(2,22) = 8.374$, $P = 0.002$] and a significant interaction of visual field and eccentricity [$F(2,22) = 6.788$, $P = 0.005$]. Post hoc Bonferroni adjusted pairwise comparisons for eccentricity showed that there were significant differences between 2° and 6.5° eccentricities ($P = 0.002$) and a trend ($0.05 < P < 0.1$) for 2° and 4.5° ($P = 0.053$). Paired-sample t -tests were carried out to assess visual-field differences between each dependent variable. For the accuracy analysis for the 4.5° eccentricity, a significant left vs. right visual-field difference was seen for the rPPC [$t(11) = 2.241$, $P = 0.047$] and rVO [$t(11) = 3.388$, $P = 0.008$] conditions (Fig. 5H). Accuracy for the most-lateral eccentricity of 6.5° showed significant left vs. right visual-field differences for the rPPC [$t(11) = -3.137$, $P = 0.023$; Fig. 5I]. The RT analysis revealed a three-way interaction of stimulation sites, visual field, and eccentricity [$F(6,60) = 2.515$, $P = 0.031$], but no two-way interaction or main effect was seen, with a trend ($0.05 < P < 0.1$) for a visual-field main effect [$F(1,10) = 4.690$, $P = 0.056$]. Consequently, no post hoc tests were carried out for this analysis.

Experiment 2

For the experiment, using a feature search task carried out for comparison with the conjunction search task, we found no significant effects of TMS delivered over rFEF, rPPC, or rVO in near or far space, as shown by no significant two-way,

repeated-measures ANOVA interaction of stimulation site and viewing distance in the d-prime analysis [$F(4,44) = 0.279$, $P = 0.890$], with no significant main effect of stimulation site [$F(4,44) = 0.665$, $P = 0.620$] or viewing distance [$F(1,11) = 0.640$, $P = 0.441$]. Similarly, no significant interaction was found in the target-present RT analysis [$F(4,44) = 0.970$, $P = 0.434$], with a trend ($0.05 < P < 0.1$) for the stimulation site [$F(4,44) = 2.343$, $P = 0.069$] and no significance of viewing distance [$F(1,11) = 1.971$, $P = 0.188$]. Interestingly, we found a significant positive-response bias (Fig. 5) all across stimulation sites in far space, shown by significant main effects of viewing distance [$F(1,11) = 5.383$, $P = 0.041$], although the interaction was not significant [$F(4,44) = 0.197$, $P = 0.939$]. This could imply that subjects tended to make more false alarms in the far viewing-distance condition, even though this bias did not affect their accuracy or RT.

DISCUSSION

TMS delivered over rPPC disrupted performance on a conjunction search task, where trials with eye movements were excluded only when it was presented in far space and with decreased accuracy for the most eccentric far-space, left visual-field targets. Disruption was also seen in far space for rVO stimulation and without an effect of distance for rFEF stimulation. These results for PPC stimulation mimic far-space neglect seen in patient-neglect studies. No effects were seen for any site with a feature search task presented in a similar manner.

The parietal component of the dorsal attention network serves as a hub for visuospatial functions across multiple cortical areas within the frontal and temporal lobes (Kravitz et al. 2011). Specifically, in patient studies, parietal-frontal white matter damage involving the anterior fascicle or the superior longitudinal fascicle results in a disconnection of large portions of brain areas, including the parietal, parietal-temporal, and temporal cortex from frontal areas, and produces neglect symptoms (Doricchia et al. 2008). It has been assumed that these parietal circuits play a specific role in spatial processing (Driver and Mattingley 1998; Driver and Vuilleumier 2001) and that the parietal cortex performs functions that are effective for target identification processes, especially in feature integration (Nobre et al. 2003).

Responses to stimulus saliency may be one function of rPPC (Mevorach et al. 2006), particularly in spatially specific aspects of top-down visual selection (Hung et al. 2005). TMS delivered over rPPC has been found to reduce interference in search from a salient singleton distractor, and this is argued to be due to elimination of top-down control (Hodsoll et al. 2009). In a neuroanatomical model, top-down control has been suggested to be the ventral frontoparietal network, functioning as a "circuit breaker" (Corbetta and Shulman 2002), directing a more dorsal, bilateral network to the presence of a salient event (Beck and Kastner 2009; Corbetta and Shulman 2002).

In addition, rPPC processes information from the extrastriate cortex to generate a response-weighted transformation into the appropriate body coordinate system required to allow action in space (Ellison et al. 2003), and these data have suggested an association with representation of the body spatiality (Muggleton et al. 2006). The neural basis of this spatial cognition comprises a number of connections between cortical and sub-

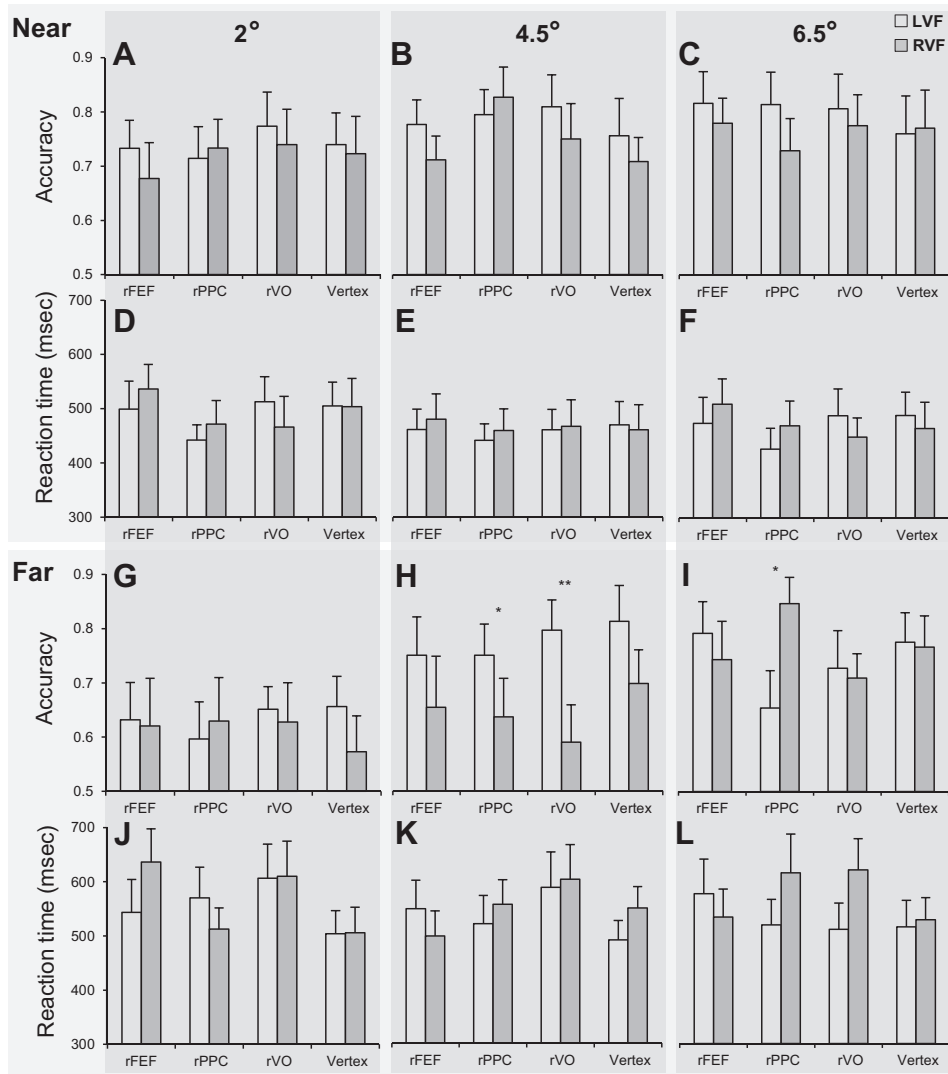


Fig. 5. The hemifield analysis. Accuracy and RT analysis of left vs. right visual fields over 3 different eccentricities from the horizontal meridian [2° , 4.5° , and 6.5° of visual angle (*experiment 1*)]. A–C: accuracy in near space for 2° , 4.5° , and 6.5° of visual angle, respectively; D–F: mean correct target-present RT in near space for 2° , 4.5° , and 6.5° of visual angle, respectively; G–I: accuracy in far space for the same eccentricities; J–L: mean correct target-present RT in far space for the same eccentricities. * $P < 0.05$; ** $P < 0.01$; error bars: SE.

cortical brain regions (Doricchia 2008; Halligan et al. 2003). The functional heterogeneity of the PPC suggests that space is not represented as a single map but instead, as an interconnection of specialized areas that plays a role in different behaviors (Duhamel et al. 1997), and evidence of PPC function in spatial navigation can be observed on patients with lesions to PPC, who can no longer orientate and navigate within space (Berti et al. 2002).

In examining the role of PPC in near- and far-space neglect, Bjoertomt et al. (2002) found that TMS over rPPC produced a leftward shift in the perceived midpoint of left elongated lines in near space. In a study of monkeys, near space seemed to be represented in the rostral part of the inferior parietal lobe (area 7b) (Leinonen et al. 1979)—evidence supported by Duhamel et al. (1997), who found that ventral intraparietal (VIP) neurons are sensitive to near or approaching visual stimuli. However, there are contradictory functions of the parietal cortex regarding near- and far-space coding. The VIP area codes specifically for near space, and interestingly, one specific area in the parietal cortex—the lateral intraparietal (LIP) area—was found to code far space (Hubbard et al. 2005). The LIP neurons in this area carry visual, memory, and saccade-related signals that describe stimuli in terms of the distance and direction of the

stimulus location and thus process the connections that provide input from distant regions of the visual field (Colby and Goldberg 1999).

Although studies have shown a near-space role for rPPC (Bjoertomt et al. 2002; Lane et al. 2013), in this current study, no effect of TMS over rPPC in the near-space condition was seen, and the effect was also seen for conjunction search and not feature search. There are several possible explanations for why this was the case. In terms of the near/far-space difference, peripheral arrangement of the stimuli in the conjunction visual search array may have meant that rPPC was more important, specifically in far space, for this type of arrangement. In this vein, Aimola et al. (2012) found that variations in task demands could affect the degree of neglect impairment in near and far space. Additionally, our TMS effect with stimulation over rPPC in far space (Fig. 4) may be consistent with the rPPC having an important role in an eye-fixed coordinate system in near and far space for representing reaching and pointing to targets (Medendorp et al. 2003). Furthermore, some patient studies also support the importance of the parietal cortex in far space. Consistent with our findings, a patient study by Cowey et al. (1994) reported that the patients with severe left visuospatial neglect showed greater errors in an angular-line bisec-

tion task for targets in far space than for those made in near space. Berti and Frassinetti (2000) also found that reaching behavior of a neglect patient with parietal damage revealed an increase in displacement error for far space.

Dysfunction of rPPC will typically result in a spatial bias toward the right visual field and a neglect of the left visual field, presumably due to a shift in weighting toward contralateral space (Szczepanski et al. 2010). Specifically, in PPC, six topographically organized areas were located along the intra-parietal sulcus (Sheremata et al. 2010; Silver and Kastner 2009; Szczepanski et al. 2010), containing a representation of the contralateral visual hemifield (Silver and Kastner 2009).

The neglect-like effect appeared only in the rPPC area for targets at 6.5° of eccentricity (Fig. 5I). This seems consistent with the findings of Rosenthal et al. (2006), who were unable to demonstrate parietal involvement in visual search when targets were presented at a fixed, small eccentricity in a circular array of diameter 2.4°. This was assumed to be due to a role of the parietal cortex in processing peripheral targets, which was seen in a macaque single-neuron recording study of area 7a, where Battaglia-Mayer et al. (2007) found that coexistence signals were observed for peripheral target visual stimuli. A macaque microstimulation study by Cutrell and Marrocco (2002) also suggested that the parietal cortex will initiate covert shifts of attention with peripheral targets. This parietal function is presumably due to the large receptive field (15–20° of visual angle) of a cell in the VIP area, as shown by macaque single-cell recording studies (e.g., Hupé et al. 2001). With the use of TMS over rPPC, Muggleton et al. (2006) found better contrast discrimination of the stimuli in the central rather than in the peripheral visual field and in a visual extinction study, and Cazzoli et al. (2009) also found that visual extinction induced by theta-burst stimulation over rPPC was more pronounced for eccentric left-sided stimuli. This rPPC function, related to peripheral targets, could be due to a role for rPPC in calculating the reference frame of each target, as well as calculating the outer limit of the target location in spatial processing (Muggleton et al. 2006; Wilson et al. 2005; Woodin and Allport 1998). It may be that the neurons in the rPPC not only code the stimulus in a simple retinotopic coordinate frame but also code several reference frames, including an object-based reference frame that is required for further actions, such as reaching and visuomotor behavior (Colby and Goldberg 1999). In the present study, where eye movements did not occur, parietal activation is required for a covert attentional search to efficiency, to locate the stimuli that are located mainly in the peripheral visual field.

Furthermore, the results showed that the accuracy was better in the left visual field across all areas in far space (Fig. 5, *H* and *J–L*), except in the case of rPPC TMS discussed above, consistent with previous studies, in which in healthy subjects show overattendance toward the left visual hemispace (“pseudoneglect”) (Giglia et al. 2011; Heber et al. 2010; Mesulam 1999). This is thought to be due to the right-hemisphere dominance for visuospatial attention, resulting in a bias of competition, producing displacement of the perceived midpoint toward one side (McCourt et al. 2005), such as that seen when using a peripheral visual conjunction search paradigm, pseudoneglect could only be observed in far space and not in near space.

In this experiment, the results over rFEF and rVO were consistent with previous research. TMS over rFEF resulted in a decrease of performance regardless of distance (Lane et al. 2013). For rFEF, this is consistent with the rFEF function in visuospatial control, which in a visual search, has been argued to be controlling spatial attention (Grosbras and Paus 2002; Juan et al. 2008), especially for visual selection (Juan et al. 2008). rVO has also been identified as important in attention to far space (Weiss et al. 2000), possibly because of greater involvement of the ventral stream in the far-space condition (Weiss et al. 2003). Furthermore, Bjoertomt et al. (2002) found that TMS over rVO produced a significant effect for bisected lines in far space. It is also important to discuss these results in the context of visual search performance, with first, neglect only seen for the conjunction search task and second, the seeming discrepancy between the results here and the relatively large amount of data showing an involvement of PPC in visual conjunction search (predominantly for near-space targets). Whereas both of these issues would certainly benefit from more investigation, there is evidence to suggest that at least in the case of mild neglect, symptoms may be more noticeable in more complex tasks. Along these lines, Taylor (2003) found that patients with mild neglect showed no effects with conventional clinical testing but did show effects when more complex tasks were used. In terms of the absence of a conjunction search disruption in near space, it has been shown previously that search performance is not affected when the need for spatial search is removed (Ellison et al. 2003). Whereas in the current study, the stimuli were quite different from Ellison et al. (2003), who used a single, central stimulus, it may be that the use of a number of fixed locations (vs. the usual random arrays used) reduced the involvement of PPC to a sufficient degree that no TMS disruption was seen.

Conclusion

The results presented here revealed rPPC involvement in search in far space by using a conjunction visual search task in elliptical peripheral array, with a pattern consistent with stimulation, resulting in neglect. This rPPC involvement in far space is different from many studies, regarding rPPC function in space, which have typically examined rPPC involvement only in near space. Further studies may provide insight into both the nature of the role of rPPC in terms of its contribution to neglect as well as differences in its involvement in tasks, such as conjunction search performance in near and far space. It seems that PPC is involved in far-space neglect when the stimuli or task are not trivial in terms of complexity, and also, the higher level of consistency of target locations reduces its role in conjunction search in general.

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DISCLOSURES

The authors declare no competing financial interests.

AUTHOR CONTRIBUTIONS

Author contributions: I.T.M., C-H.J., and N.G.M. conception and design of research; I.T.M., C-L.L., and C.F.C. performed experiments; I.T.M., C-L.L.,

and N.G.M. analyzed data; I.T.M., C-H.J., and N.G.M. interpreted results of experiments; I.T.M. and N.G.M. prepared figures; I.T.M. and N.G.M. drafted manuscript; I.T.M. and N.G.M. edited and revised manuscript; I.T.M., D.L.H., O.J.L.T., C-H.J., and N.G.M. approved final version of manuscript.

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